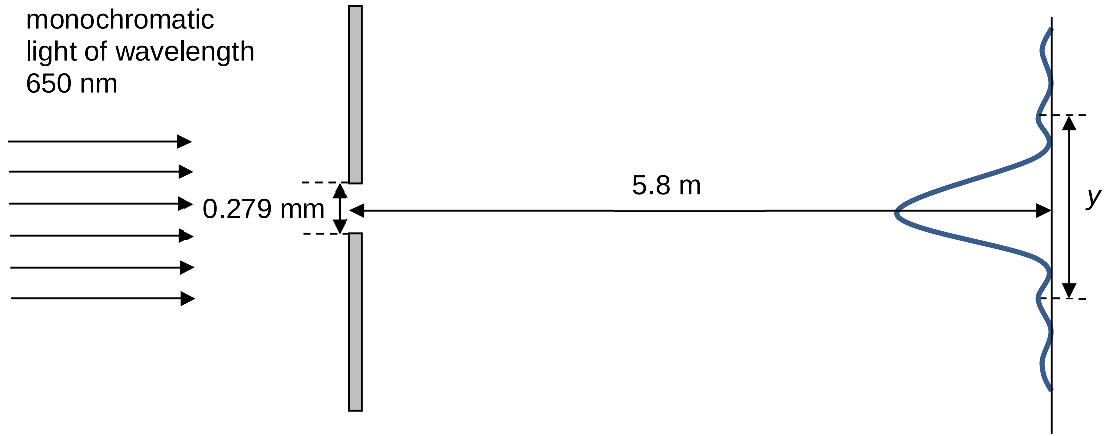
20 A laser light of wavelength 650 nm is passed normally through a narrow slit. A screen is placed parallel to the slit 5.8 m away from the slit. An interference pattern is formed on the screen. The width of the slit is 0.279 mm. The distance *y* is the distance between the two first maximas.



What is the value of *y*?

A 13.5 mm **B** 27.0 mm **C** 40.5 mm **D** 54.0 mm

Ans: C
Since $\sin\theta = \lambda/b$
$\theta = 0.1333°$
Distance from central maxima to first minima = 13.5 mm
y = 27 + 13.5 = 40.5 mm
(since width of subsequent maxima is half the width of central maxima)

21 A negatively charged sphere P is balanced halfway between two horizontal plates when a